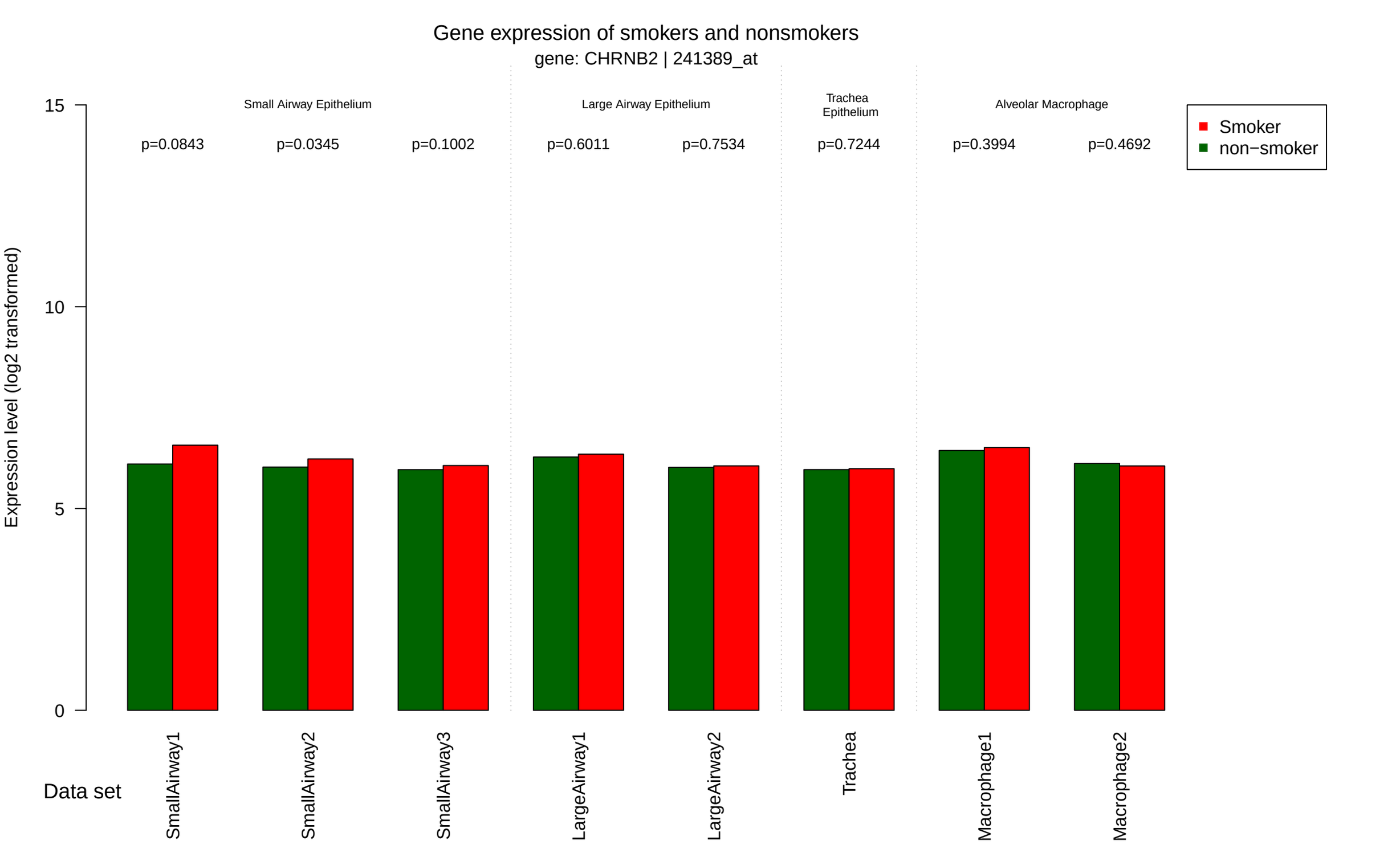
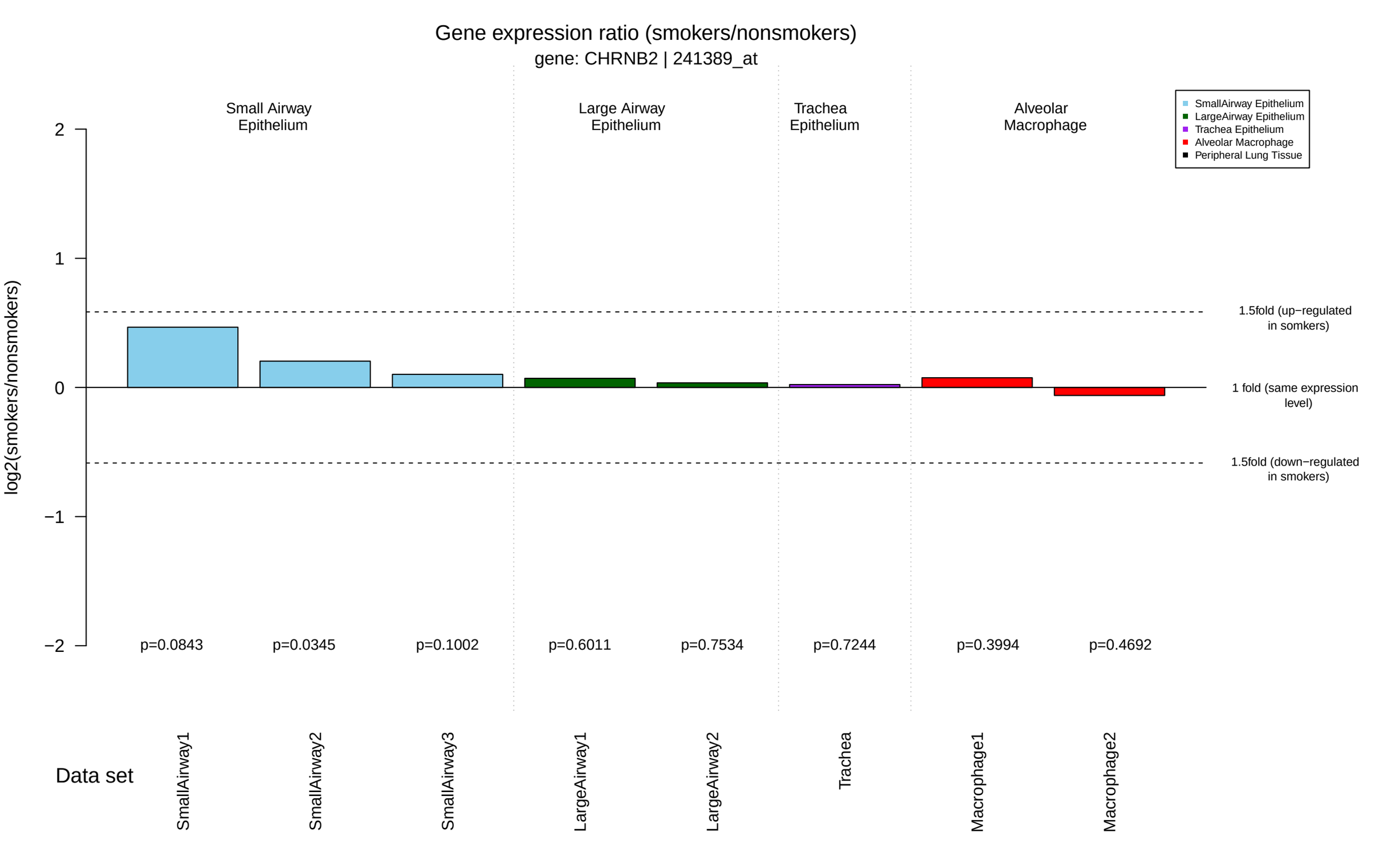
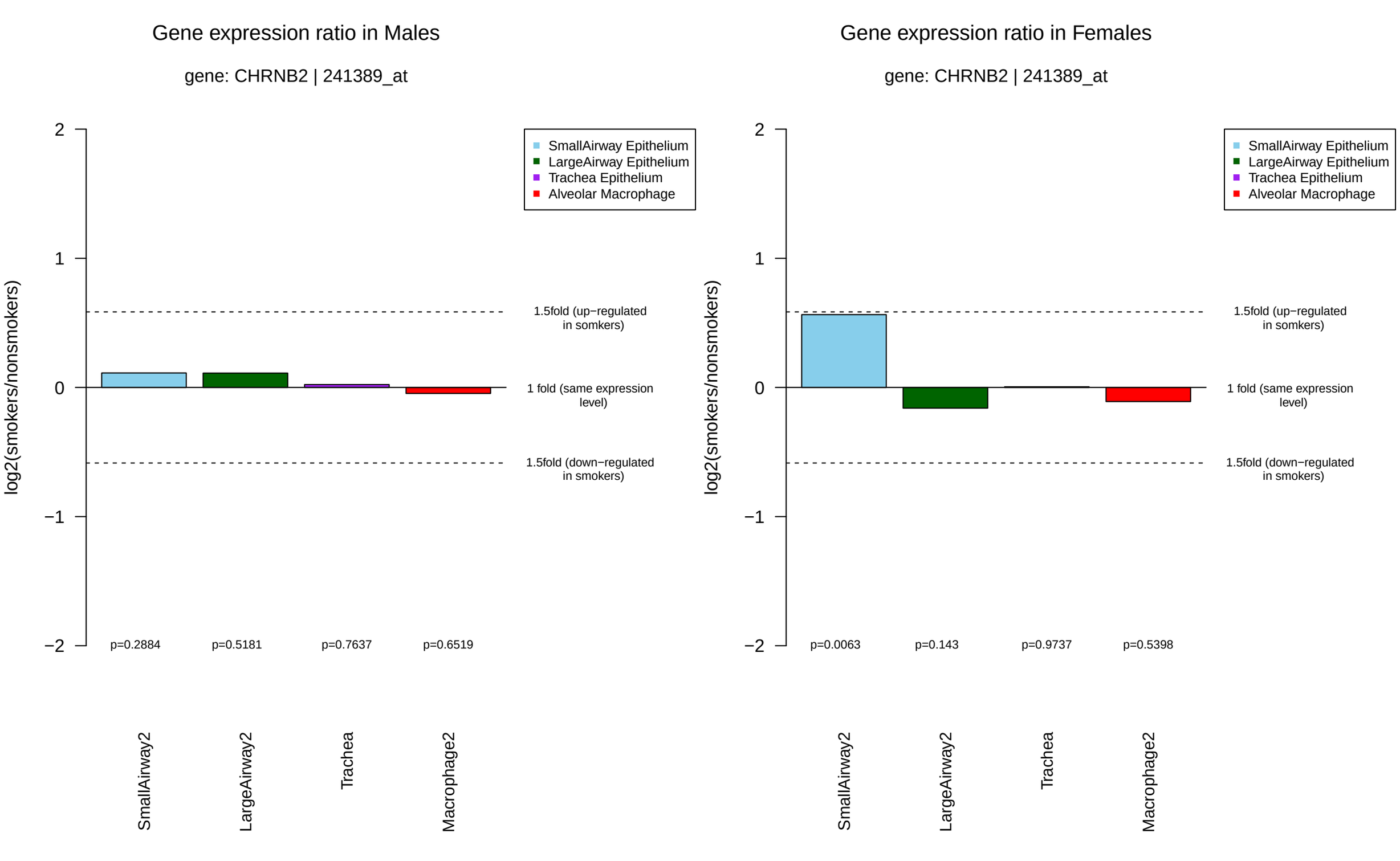
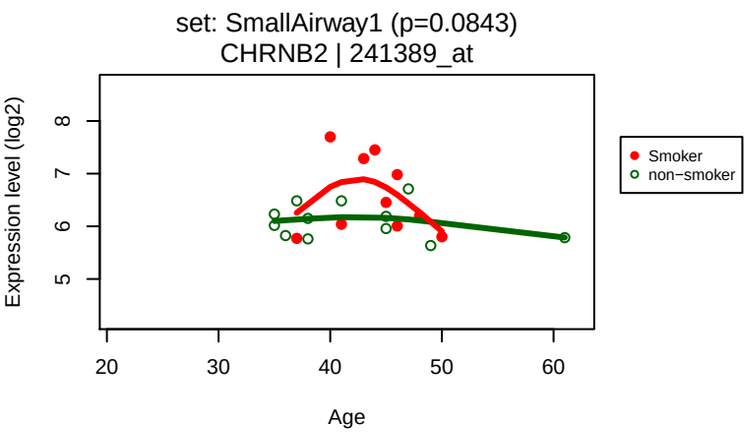
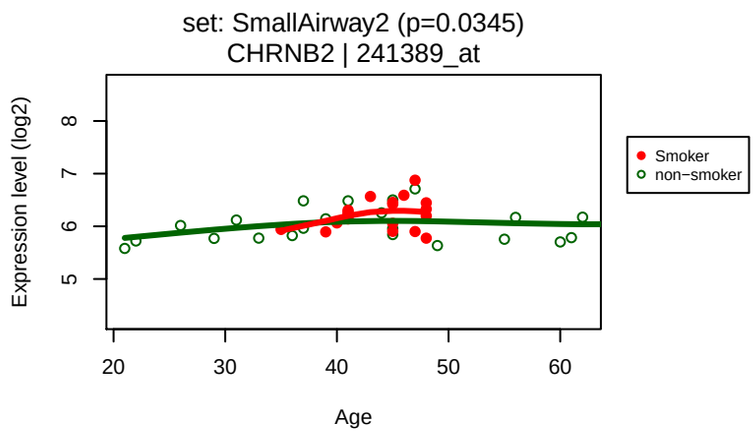
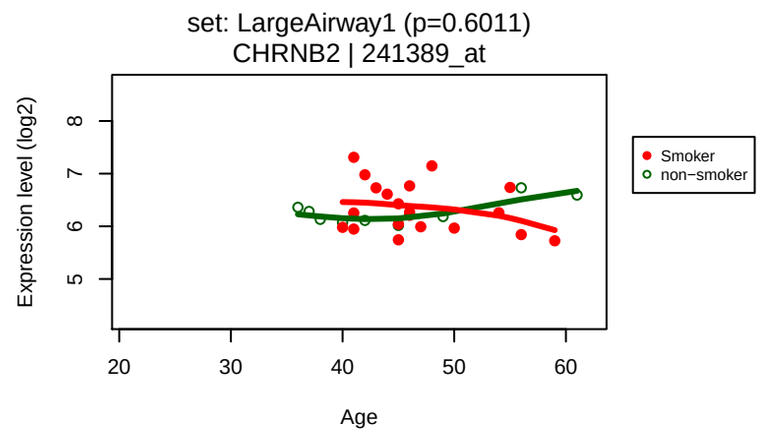

Gene expression of smokers and nonsmokers
gene: CHRNB2 | 241389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0843
p=0.0345
p=0.1002
p=0.6011
p=0.7534
p=0.7244
p=0.3994
p=0.4692
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

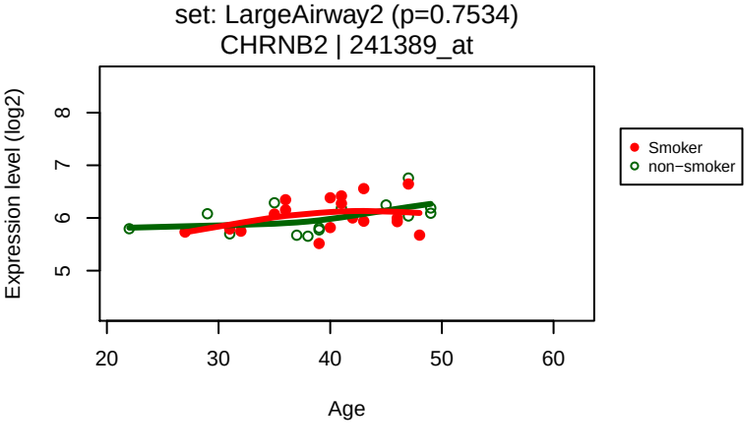

Gene expression ratio (smokers/nonsmokers)
gene: CHRNB2 | 241389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0843
p=0.0345
p=0.1002
p=0.6011
p=0.7534
p=0.7244
p=0.3994
p=0.4692
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

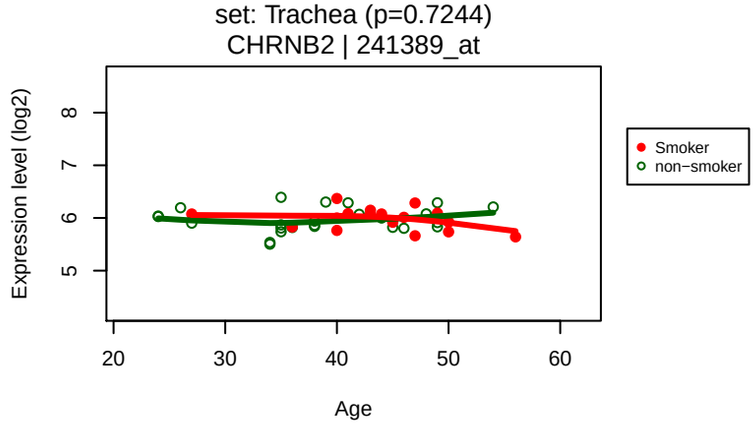

Gene expression ratio in Males
gene: CHRNB2 | 241389_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2884
p=0.5181
p=0.7637
p=0.6519
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNB2 | 241389_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0063
p=0.143
p=0.9737
p=0.5398
SmallAirway2
LargeAirway2
Trachea
Macrophage2

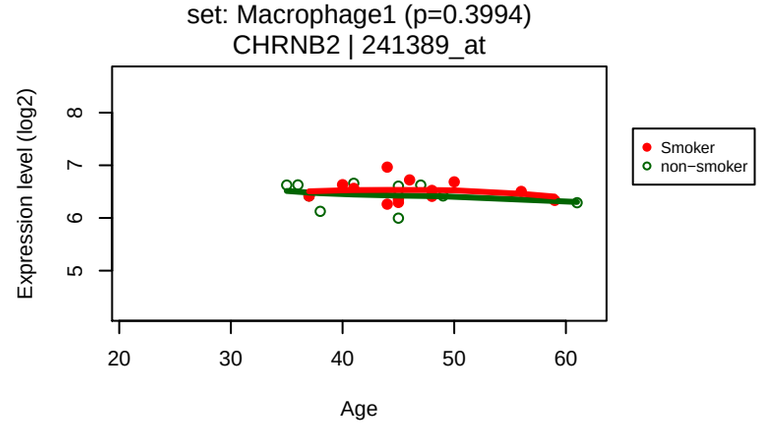

set: SmallAirway1 (p=0.0843)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker

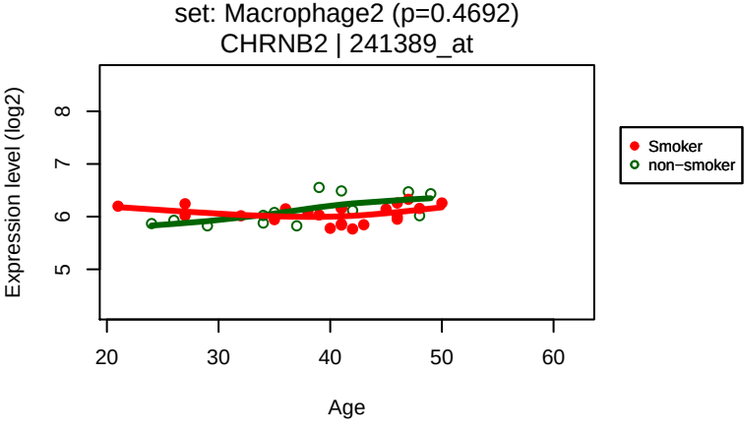

set: SmallAirway2 (p=0.0345)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker

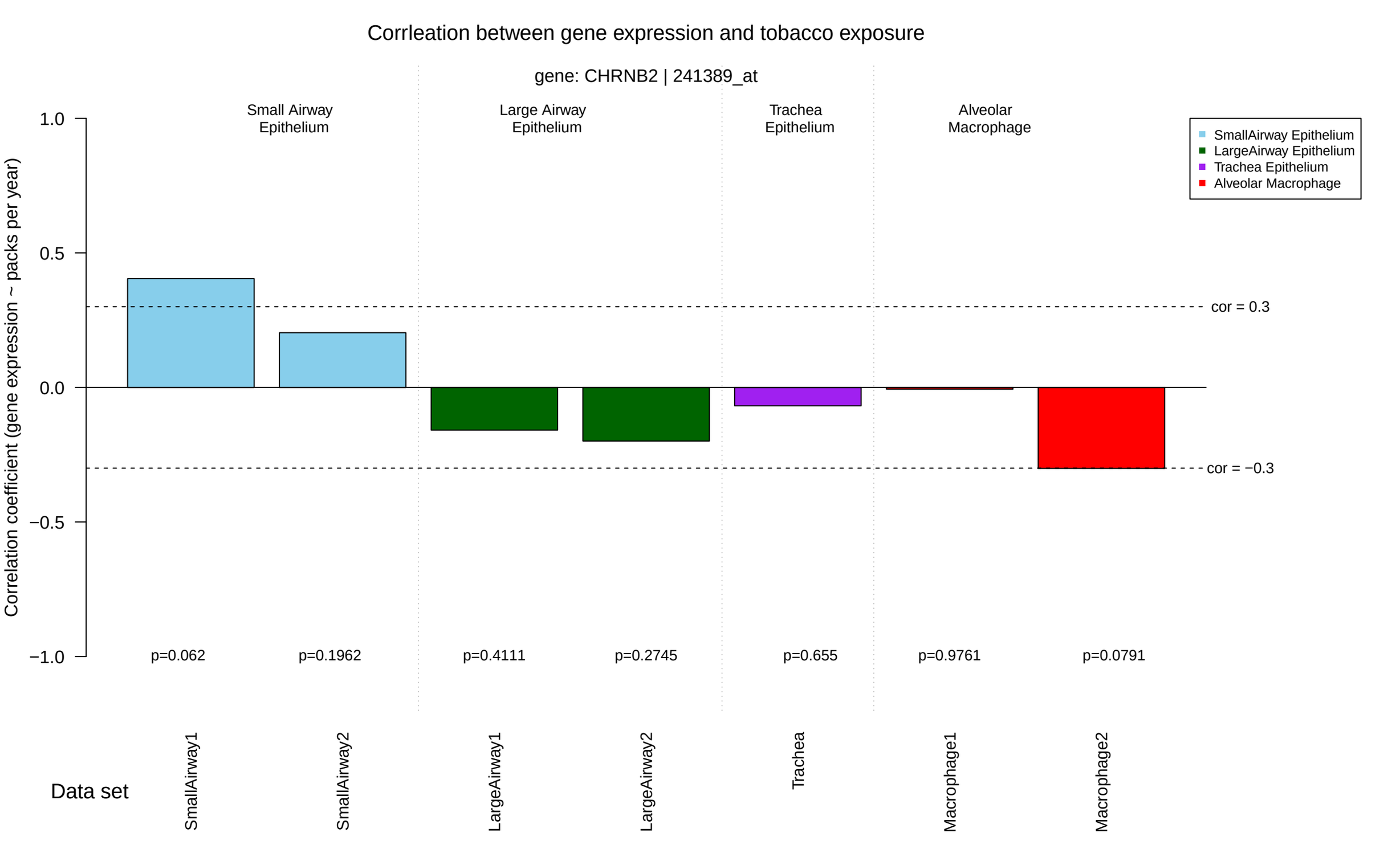

set: LargeAirway1 (p=0.6011)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7534)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7244)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3994)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4692)
CHRNB2 | 241389_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNB2 | 241389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.062
p=0.1962
p=0.4111
p=0.2745
p=0.655
p=0.9761
p=0.0791
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2